Round-3 TNC Fit (2026-06-23): Fission + (n,tot) Data Revision

Comprehensive fission and total-cross-section data review by R. Capote and I. Durán (2026-06-23 discussions), built on the round-2 Step-5 favoured fit. Audit trail: `references/emails/tnc_update_roberto_20260623/`*, with a datapoint-level before→after manifest in* `data_change_manifest.md`*.*

This round applies a large fission-data extension (new absolute and ratio measurements across all four fissile nuclides), value/uncertainty updates to the total-cross-section (transmission) entries from local EXFOR re-fits, removal of identified outliers and duplicates, and an update of the U-234(n,γ) value to the B80 evaluation. The **current SCA/SCR classification of every existing entry is kept** per the instruction; only values and uncertainties change on updated entries. The fit configuration is identical to round-2 Step 5 (30 free parameters; Lounsbury alpha included; the four Lemmel SCA–SCR points retained, so all four SCR parameters remain in the fit).

Every interpretation that departs from a literal reading of the source tables is recorded in the decisions appendix (flags F1–F13) for review.

Overview

Fit	N meas	N free	DOF	χ^2	χ^2/DOF	Converged
Round-2 Step 5 (base)	127	30	97	80.55	0.830	Yes
Round 3 (2026-06-23)	**143**	30	113	**73.42**	0.650	Yes

Full database: 205 measurements (189 base − 10 deletions + 26 additions); 143 enter the microscopic fit after the Maxwellian filter. χ^2 *decreases* despite the net +16 measurements, because the deletions remove high-residual outliers while the new fission data are mutually consistent.

Change Summary

Category	Count	Serial numbers
Transmission value/σ updates	15	63, 64, 65, 73, 74, 75, 76, 78, 85, 87, 186, 187, 188, 189, 193
Fission value/σ updates	2	98, 108
U-234(n,γ) ⟶ 100.9 b (B80)	1	5
Deletions (outliers/duplicates)	10	71, 79, 86, 103, 104, 106, 107, 190, 191, 194
Additions (23 fission + 3 transmission)	26	195–220

Discarded and Excluded Datapoints

Because the input-data list above shows only what remains in the database, this section makes the removals explicit. **Part 1** lists the entries deleted from the Step-5 base (their pre-deletion values); **Part 2** lists candidates that were considered in the 2026-06-23 review but not entered.

Deleted from the database (10 entries)

No.	Author	Reference	Quantity	Value	σ (abs)	σ (%)	Reason
71	GERASIMOV	1962(1)	ABS 35	670.31	14	2.089	Duplicate of the total-XS entry No. 186 (bare ABS is a derived quantity from the same experiment).
79	PATTENDEN	1956(1)	(ABS 33)+SCR 33	596.27	18	3.019	U-233 (n,tot): discarded — too much scatter.
86	CRAIG	1964(1)	(ABS 41)+SCA 41	1380.6	40	2.897	Pu-241 (n,tot): discarded — too large scattering (see plot).
103	RAFFLE	1959(1)	FIS 35	581.97	18	3.092	U-235 fission: outlier.
104	RAFFLE	1959(1)	39 FH1 39	1714.8	48.8	2.843	Pu-239 fission: outlier.

No.	Author	Reference	Quantity	Value	σ (abs)	σ (%)	Reason
106	MASLIN	1965(1)	FIS 35	583.71	8.46	1.449	U-235 fission: outlier.
107	SAPLAKOGLU	1959(1)	FIS 35	593.17	13.2	2.232	U-235 fission: outlier.
190	HARVEY-2	Capote 2026-05-04 list (1979)	(ABS 33)+SCR 33	592	6	1.014	U-233 (n,tot): second Harvey 79 entry removed; the surviving Harvey is No. 189, updated to 588(9) [F10].
191	SCHWARTZ	Capote 2026-05-04 list (1958)	(ABS 41)+SCR 41	1410	80	5.674	Pu-241 (n,tot): discarded — too large scattering (see plot).
194	LEONARD	Capote 2026-05-04 list (1956)	(ABS 39)+SCR 39	1055	13	1.232	Pu-239 (n,tot): outlier [F2] (the (n,tot) table showed an adjusted 1042(25), but it is flagged an outlier and removed).

Considered but not added

Author	Channel	Value	Reason
Berceanu	U-235 fission	589.3 (0.8%)	Superseded by Borcea 73 = 586.3 (1.3%), per "replace by Borcea 73" (No. 212) [F9].
Bollinger 52	U-233 (n,tot)	592.6 (20 b)	In the local working notes only; omitted from the consolidated U-233 (n,tot) list [F4].
Havens	Pu-239 (n,tot)	~1080 b	Mis-calibrated experiment / outlier — "do not consider".
Bowman, Melkonian, Schrack, Wagemans 79/88, Mihailescu	fission	—	Shape measurements (relative to the thermal point); not entered as absolute datapoints [F12]. (This Melkonian ≠ the U-235 (n,tot) No. 63, which is retained.)

Fitted Quantities (Round 3 vs Step-5 base)

Shifts exceeding 0.1% relative to the Step-5 base are highlighted in **bold**. K1 = F3ETA = NUB·FF − FA is the central derived quantity.

Quantity	Step 5	Round 3	Unc (%)	Δ vs Step 5 (%)
SCA 33	13.3908	**13.3371**	2.615	**-0.401**
SCA 35	14.1213	14.1157	1.264	-0.040
SCA 39	7.4728	**7.4805**	5.105	**+0.104**
SCA 41	11.9951	11.9908	6.657	-0.036
SCR 33	12.3795	**12.3135**	3.759	**-0.533**
SCR 35	12.7734	12.7656	2.628	-0.061
SCR 39	6.6595	**6.7373**	15.929	**+1.169**
SCR 41	10.8374	**10.9254**	24.836	**+0.812**
ABS 33	573.4874	**574.9932**	0.239	**+0.263**
ABS 35	683.4317	**684.4781**	0.144	**+0.153**
ABS 39	1020.6859	**1022.0765**	0.290	**+0.136**
ABS 41	1383.6990	**1380.6308**	0.436	**-0.222**
FIS 33	528.6252	**530.3673**	0.255	**+0.330**
FIS 35	584.3285	**585.4044**	0.141	**+0.184**
FIS 39	749.1488	**750.0765**	0.238	**+0.124**
FIS 41	1018.5549	**1016.6900**	0.514	**-0.183**
CA 33	44.8622	**44.6259**	1.785	**-0.527**
CA 35	99.1032	99.0737	0.767	-0.030

Quantity	Step 5	Round 3	Unc (%)	Δ vs Step 5 (%)
CA 39	271.5372	**272.0000**	0.884	**+0.170**
CA 41	365.1441	**363.9407**	1.251	**-0.330**
CAP 34	95.9000	**100.9000**	2.086	**+5.214**
NUB 33	2.4914	2.4908	0.183	-0.025
NUB 35	2.4304	2.4301	0.168	-0.013
NUB 39	2.8824	2.8823	0.190	-0.005
NUB 41	2.9444	2.9440	0.206	-0.013
NUB 52	3.7668	3.7666	0.129	-0.004
ETA 33	2.2965	2.2975	0.197	+0.042
ETA 35	2.0780	2.0783	0.186	+0.018
ETA 39	2.1156	2.1152	0.261	-0.017
ETA 41	2.1674	2.1680	0.370	+0.026
ALPHA 33	0.0849	**0.0841**	1.869	**-0.853**
ALPHA 35	0.1696	**0.1692**	0.811	**-0.213**
ALPHA 39	0.3625	0.3626	0.919	+0.047
ALPHA 41	0.3585	**0.3580**	1.466	**-0.147**
WGA 33	1.0000	1.0000	0.100	+0.000
WGA 35	0.9820	0.9820	0.305	+0.000
WGA 39	1.0780	1.0780	0.928	+0.000
WGA 41	1.0460	1.0460	0.669	+0.000
WGF 33	0.9970	0.9970	0.100	+0.000
WGF 35	0.9810	0.9810	0.510	+0.000
WGF 39	1.0520	1.0520	1.331	+0.000
WGF 41	1.0520	1.0520	0.665	+0.000
FA 33	573.4874	**574.9932**	0.259	**+0.263**
FA 35	671.1299	**672.1575**	0.338	**+0.153**
FA 39	1100.2995	**1101.7984**	0.972	**+0.136**
FA 41	1447.3492	**1444.1398**	0.799	**-0.222**
FF 33	527.0393	**528.7762**	0.274	**+0.330**
FF 35	573.2262	**574.2817**	0.529	**+0.184**
FF 39	788.1045	**789.0804**	1.352	**+0.124**
FF 41	1071.5198	**1069.5579**	0.841	**-0.183**
F1ETA 33	2.2897	2.2906	0.243	+0.042
F1ETA 35	2.0758	2.0762	0.623	+0.018
F1ETA 39	2.0646	2.0642	1.643	-0.017
F1ETA 41	2.1798	2.1804	1.014	+0.026
F2ETA 33	1313.0907	**1317.0883**	0.312	**+0.304**
F2ETA 35	1393.1589	**1395.5398**	0.552	**+0.171**
F2ETA 39	2271.6351	**2274.3355**	1.363	**+0.119**
F2ETA 41	3154.9640	**3148.7864**	0.841	**-0.196**
F3ETA 33	739.6033	**742.0951**	0.450	**+0.337**
F3ETA 35	722.0290	**723.3823**	1.085	**+0.187**
F3ETA 39	1171.3357	**1172.5370**	2.763	**+0.103**
F3ETA 41	1707.6149	**1704.6467**	1.541	**-0.174**

K1 = F3ETA (Round 3)

Nuclide	K1 = F3ETA	Unc (%)	Step-5 base	Δ%
^{233}U	742.10	0.450	739.60	+0.337%
^{235}U	723.38	1.085	722.03	+0.187%
^{239}Pu	1172.54	2.763	1171.34	+0.103%
^{241}Pu	1704.65	1.541	1707.61	-0.174%

K1(^{235}U) = 723.4, consistent with the value (~722.4) anticipated in the 2026-05-06 discussion using the same B81-derived Westcott factor. The new fission data shift all four K1 values by less than 0.4%.

Full Input-Data List (absolute + relative uncertainty)

Per the 2026-06-23 "new TNC fit" request, the complete round-3 input database with **both absolute and relative uncertainty** for every measurement, for cross-checking against Axton and the 2026-06-23 changes. Status: **NEW** = added this round (Nos. 195–220); **UPD** = value/uncertainty updated this round; blank = unchanged from the Step-5 base. "In fit" marks the measurements that pass the microscopic filter and enter the GLS fit (others are Maxwellian-derived and used only for derived-quantity folding).

No.	Author	Reference	Quantity	Value	σ (abs)	σ (%)	Status	In fit
1	LAPONCHE	1972(2)	(FA 39)/FA 35	1.6491	0.00961	0.583		
2	CORNISH	1956(2)	CAP 39	312	14	4.490		
3	HALPERIN	1963(2)	CAP 33	49.53	3.17	6.400		
4	CABELL	1971(2)	FLEM	0.09827	0.00329	3.348		
5	LEMMEL	1982(1)	CAP 34	100.9	2.1	2.086	UPD	yes
6	POPOVIC	1953(2)	FF 35	571.1	12.6	2.211		
7	POPOVIC	1955(2)	33 FFH 33	844.72	26.6	3.147		
8	KEITH	1968(2)	33 FFH 33	834.15	11.4	1.371		
9	KEITH	1968(2)	33 FFH 35	889.01	12.9	1.446		
10	KEITH	1968(2)	39 FFH 39	1884.4	17.9	0.949		
11	BIGHAM	1975(2)	33 FFH 33	835.2	5.5	0.659		
12	JAFFEY	1955(2)	(FF 41)/FF 39	1.3552	0.0193	1.424		
13	WHITE	1967(2)	(FF 39)/FF 35	1.3583	0.0291	2.139		
14	WHITE	1967(2)	(FF 41)/FF 35	1.8806	0.0527	2.802		
15	VIDAL	1970(2)	(FF 33)/FF 35	0.932	0.009	0.966		
16	BIGHAM	1975(2)	(FF 33)/FF 35	0.9323	0.0042	0.450		
17	SWEET	1973(2)	(39 FFH 39)/34 FFH 35	1.3544	0.0192	1.414		
18	BIGHAM	1975(2)	(39 FFH 39)/33 FFH 33	2.2796	0.0153	0.669		
19	BIGHAM	1975(2)	(39 FFH 39)/33 FFH 35	2.1239	0.0151	0.709		
20	BIGHAM	1975(2)	F1BIG	0.5527	0.00311	0.562		
21	LOUNSBURY	1970(1)	(CA 33)/FIS 33	0.0861	0.00193	2.239		yes
22	LOUNSBURY	1970(1)	(CA 35)/FIS 35	0.1697	0.0029	1.709		yes
23	LOUNSBURY	1970(1)	(CA 39)/FIS 39	0.3555	0.0057	1.603		yes
24	INGHRAM	1956(2)	(CAP 33)/FF 33	0.09403	0.003	3.190		
25	CORNISH	1960(2)	(CAP 35)/FF 35	0.188	0.014	7.447		
26	OKAZAKI	1964(2)	(CAP 33)/FF 33	0.0902	0.00156	1.731		
27	OKAZAKI	A1964(2)	(CAP 35)/FF 35	0.1705	0.0018	1.056		
28	LISMAN	1967(2)	(CAP 33)/FF 33	0.093	0.00198	2.128		
29	LISMAN	1967(2)	(CAP 35)/FF 35	0.1712	0.00152	0.886		
30	CONWAY	1967(2)	(CAP 33)/FF 33	0.0851	0.0042	4.935		
31	CONWAY	1967(2)	(CAP 35)/FF 35	0.1705	0.0072	4.223		
32	DURHAM	1967(2)	(CAP 35)/FF 35	0.1746	0.002	1.145		
33	DURHAM	1967(2)	(CAP 39)/FF 39	0.3882	0.0059	1.520		
34	CABELL	1971(2)	(CAP 33)/FF 33	0.0857	0.00261	3.048		
35	CABELL	1971(2)	(CAP 35)/FF 35	0.1696	0.00298	1.759		
36	CABELL	1971(2)	(CAP 39)/FF 39	0.3812	0.0183	4.808		
37	DEBOISBLANC	1961(2)	(F1ETA 33)/F1ETA 35	1.115	0.00799	0.717		
38	FAST	1960(2)	(F1ETA 39)/F1ETA 35	0.997	0.0162	1.625		
39	FAST	1960(2)	(F1ETA 41)/F1ETA 35	1.0528	0.0174	1.653		
40	CABELL	1965(2)	(F3ETA 33)/F3ETA 35	1.0461	0.0247	2.357		
41	CABELL	1965(2)	(F3ETA 39)/F3ETA 35	1.6079	0.0386	2.403		
42	CABELL	1965(2)	(F3ETA 33)/F3ETA 35	1.0309	0.0404	3.919		
43	CABELL	1965(2)	(F3ETA 39)/F3ETA 35	1.6223	0.0675	4.161		
44	GWIN	1962(2)	(F2ETA 33)/F2ETA 35	0.953	0.0135	1.414		
45	GWIN	1962(2)	(F2ETA 39)/F2ETA 35	1.631	0.0231	1.414		
46	GWIN	A1962(2)	F3ETA 33	744.7	4	0.537		
47	GWIN	A1962(2)	F3ETA 35	722.7	3.9	0.540		
48	LAPONCHE	1972(2)	(F3ETA 39)/F3ETA 35	1.6058	0.0178	1.107		
49	LAPONCHE	1972(2)	(F3ETA 41)/F3ETA 39	1.4687	0.0526	3.580		
50	ALIKHANOV	1956(2)	(F3ETA 33)/F3ETA 35	1.039	0.037	3.561		
51	ALIKHANOV	1956(2)	(F3ETA 39)/F3ETA 35	1.565	0.105	6.709		
52	MUELHAUS	1959(2)	(F3ETA 33)/F3ETA 35	1.0221	0.0185	1.811		
53	MUELHAUS	1959(2)	(F3ETA 39)/F3ETA 35	1.6043	0.0364	2.271		
54	DIVADEENAM	1984(2)	WGA 33	1	0.001	0.100		yes
55	DIVADEENAM	1984(2)	WGA 35	0.982	0.003	0.305		yes

No.	Author	Reference	Quantity	Value	σ (abs)	σ (%)	Status	In fit
56	DIVADEENAM	1984(2)	WGA 39	1.078	0.01	0.928		yes
57	DIVADEENAM	1984(2)	WGA 41	1.046	0.007	0.669		yes
58	DIVADEENAM	1984(2)	WGF 33	0.997	0.001	0.100		yes
59	DERUYTTER	1985(3)	WGF 35	0.981	0.005	0.510		yes
60	DIVADEENAM	1984(2)	WGF 39	1.052	0.014	1.331		yes
61	DIVADEENAM	1984(2)	WGF 41	1.052	0.007	0.665		yes
62	EGELSTAFF	1954(1)	(ABS 35)+SCR 35	724.33	26	3.590		yes
63	MELKONIAN	1953(1)	(ABS 35)+SCR 35	699	10	1.431	UPD	yes
64	PALEVSKY	1954(1)	(ABS 35)+SCR 35	700	5	0.714	UPD	yes
65	NITIKIN	1955(1)	(ABS 35)+SCA 35	707	10	1.414	UPD	yes
66	SIMPSON	1960(1)	(ABS 35)+SCR 35	690.32	9.6	1.391		yes
67	SAFFORD	1959(1)	(ABS 35)+SCA 35	696.32	2.5	0.359		yes
68	SAFFORD	1959(1)	(ABS 35)+SCA 35	698.32	5.1	0.730		yes
69	BLOCK	1960(1)	(ABS 35)+SCR 35	693.32	5	0.721		yes
70	SAPLAKOGLU	1961(1)	(ABS 35)+SCR 35	696.32	2.5	0.359		yes
72	ZIMMERMAN	1955(1)	(ABS 39)+SCR 39	1022.5	13	1.271		yes
73	NITIKIN	1955(1)	(ABS 39)+SCA 39	1030	20	1.942	UPD	yes
74	BOLLINGER	1958(1)	(ABS 39)+SCR 39	1029	10	0.972	UPD	yes
75	PATTENDEN	A1956(1)	(ABS 39)+SCR 39	1039	25	2.406	UPD	yes
76	SAFFORD	1961(1)	(ABS 39)+SCR 39	1030	15	1.456	UPD	yes
77	MUETHER	1954(1)	(ABS 33)+SCR 33	597.27	14	2.344		yes
78	NITIKIN	1955(1)	(ABS 33)+SCA 33	580	20	3.448	UPD	yes
80	GREEN	1957(1)	ABS 33	574.26	20	3.483		yes
81	SIMPSON	1960(1)	(ABS 33)+SCR 33	587.27	4.7	0.800		yes
82	SAFFORD	1960(1)	(ABS 33)+SCA 33	585.77	5.8	0.990		yes
83	SAFFORD	1960(1)	(ABS 33)+SCA 33	585.67	2.4	0.410		yes
84	BLOCK	1960(1)	(ABS 33)+SCR 33	587.27	3	0.511		yes
85	SIMPSON	1961(1)	(ABS 41)+SCA 41	1385	15	1.083	UPD	yes
87	SMITH	1984(3)	(ABS 41)+SCR 41	1395	10	0.717	UPD	yes
89	LEMMEL	1982(1)	(SCA 33)-SCR 33	1.2006	0.6	49.977		yes
91	LEMMEL	1982(1)	(SCA 35)-SCR 35	1.7008	0.75	44.097		yes
93	LEMMEL	1982(1)	(SCA 39)-SCR 39	0.80037	1.1	137.437		yes
95	LEMMEL	1982(1)	(SCA 41)-SCR 41	1.2006	3	249.886		yes
96	DERUYTTER	1974(1)	39 FH1 39	1812.1	8.23	0.454		yes
97	DERUYTTER	1973(1)	34 FH1 35	1439.1	6.4	0.445		yes
98	DERUYTTER	1961(1)	FIS 35	587	5.87	1.000	UPD	yes
99	WHITE	1967(1)	(FIS 39)/FIS 35	1.253	0.0226	1.800		yes
100	WHITE	1967(1)	(FIS 41)/FIS 35	1.763	0.0485	2.751		yes
101	BERCEANO	1977(1)	34 FH1 35	1443	11.4	0.791		yes
102	RAFFLE	1959(1)	33 FH1 33	834.26	27.5	3.302		yes
105	RAFFLE	1956(1)	(FIS 41)/FIS 39	1.3319	0.0799	6.000		yes
108	WATANABE	1964(1)	FIS 41	1019.3	24.5	2.400	UPD	yes
109	FRAYSSE	1965(1)	(39 FH1 39)/34 FH1 35	1.2425	0.0193	1.557		yes
110	BORCEA	1970(1)	(39 FH1 39)/34 FH1 35	1.2578	0.0204	1.624		yes
111	BOLDEMAN	1977(1)	NUB 52	3.7549	0.0162	0.431		yes

No.	Author	Reference	Quantity	Value	σ (abs)	σ (%)	Status	In fit
112	BOLDEMAN	1980(1)	(NUB 33)/NUB 52	0.66049	0.00168	0.255		yes
113	BOLDEMAN	1980(1)	(NUB 35)/NUB 52	0.64318	0.00164	0.255		yes
114	BOLDEMAN	1980(1)	(NUB 39)/NUB 52	0.76614	0.00186	0.243		yes
115	BOLDEMAN	1980(1)	(NUB 41)/NUB 52	0.78028	0.0018	0.231		yes
116	SPENCER	1982(1)	NUB 52	3.7831	0.00836	0.221		yes
117	GWIN	1984(1)	(NUB 33)/NUB 52	0.65996	0.00224	0.340		yes
118	GWIN	1984(1)	(NUB 35)/NUB 52	0.64701	0.00171	0.265		yes
119	GWIN	1984(1)	(NUB 39)/NUB 52	0.7653	0.00197	0.257		yes
120	GWIN	1984(1)	(NUB 41)/NUB 52	0.78423	0.00227	0.290		yes
121	HOPKINS	1963(1)	NUB 52	3.7767	0.0316	0.838		yes
122	HOPKINS	1963(1)	(NUB 33)/NUB 52	0.65483	0.00586	0.895		yes
123	HOPKINS	1963(1)	(NUB 35)/NUB 52	0.64587	0.00541	0.838		yes
124	HOPKINS	1963(1)	(NUB 39)/NUB 52	0.75047	0.00747	0.996		yes
125	ASPLUND	1963(1)	NUB 52	3.791	0.0404	1.066		yes
126	CONDE	1965(1)	(NUB 35)/NUB 52	0.64222	0.00544	0.847		yes
127	WHITE	1968(1)	NUB 52	3.8194	0.0395	1.033		yes
128	AXTON	A1985(3)	NUB 52	3.7547	0.0113	0.300		yes
129	COLV/AXT	1966(1)	NUB 52	3.7299	0.0301	0.806		yes
130	COLV/ULL	1965(1)	NUB 52	3.7405	0.0164	0.438		yes
131	COLVIN	1965(1)	(NUB 33)/NUB 35	1.0204	0.00599	0.587		yes
132	COLVIN	1965(1)	(NUB 39)/NUB 35	1.1835	0.00799	0.675		yes
133	COLVIN	1965(1)	(NUB 41)/NUB 35	1.2102	0.011	0.905		yes
134	COLVIN	1965(1)	(NUB 35)/NUB 52	0.64334	0.00304	0.472		yes
135	MATHER	1964(1)	(NUB 33)/NUB 52	0.67093	0.00829	1.235		yes
136	MATHER	1964(1)	(NUB 35)/NUB 52	0.64145	0.00344	0.537		yes
137	MATHER	1964(1)	(NUB 39)/NUB 52	0.77307	0.00867	1.122		yes
138	ALEK-ROV	1981(1)	NUB 52	3.7618	0.0182	0.483		yes
139	SMITH	1984(1)	NUB 52	3.7678	0.0114	0.303		yes
140	EDWARDS	1982(1)	NUB 52	3.7641	0.0268	0.711		yes
141	BOZ-NESH	1977(1)	NUB 52	3.7475	0.0217	0.580		yes
142	DEVOLPI	1972(1)	NUB 52	3.7507	0.0174	0.463		yes
143	ZHANG	1981(1)	NUB 52	3.7534	0.0184	0.490		yes
144	SPIEGEL	1981(1)	NUB 52	3.7828	0.0287	0.759		yes
145	SMITH	1984(1)	ETA 33	2.2972	0.00804	0.350		yes
146	SMITH	1984(1)	ETA 35	2.0842	0.00805	0.386		yes
147	SMITH	1984(1)	ETA 39	2.112	0.00803	0.380		yes
148	SMITH	1984(1)	ETA 41	2.169	0.00909	0.419		yes
149	MACKLIN	1960(1)	ETA 33	2.3121	0.00867	0.375		yes
150	MACKLIN	1960(1)	ETA 35	2.0757	0.00793	0.382		yes
151	MACKLIN	1962(1)	ETA 39	2.1163	0.018	0.851		yes
152	CABELL	1968(1)	F1CAB	273.84	13.6	4.960		
153	CABELL	1968(1)	F2CAB	1171.9	37.2	3.176		
154	CABELL	1968(1)	F3CAB	3.269	0.0907	2.776		
155	CABELL	1968(1)	F4CAB	303.39	17.7	5.836		
156	CABELL	1968(1)	F5CAB	3.871	0.062	1.602		
157	BNL 325	1973(1)	CA 40	289.5	1.4	0.484		
158	BNL 325	1973(1)	CA 42	18.5	0.4	2.162		
159	BNL325 + WESTCOTT	1973(1)	GC116 39	1.3265	0.0803	6.056		
160	WESTCOTT	1960(1)	GC116 40	1.086	0.0217	2.000		
161	BNL325 + WESTCOTT	1973(1)	GC116 41	1.1085	0.0272	2.457		
162	WESTCOTT	1960(1)	GC116 42	1.1335	0.0303	2.670		

No.	Author	Reference	Quantity	Value	σ (abs)	σ (%)	Status	In fit
163	BNL325 + WESTCOTT	1973(1)	GA116 39	1.1846	0.018	1.521		
164	WESTCOTT	1960(1)	GA116 41	1.1073	0.00907	0.819		
165	LLOYD	1982(1)	(ETA 39)/FIS 39	0.002782	5.76e-05	2.072		yes
166	SPENCER	1985(3)	(ABS 35)+SCR 35	690	6.9	1.000		yes
167	SPENCER	1985(3)	(ABS 39)+SCR 39	1025.8	10.3	1.000		yes
168	ADAMCHUK	At. En. (R), 65, 434 (1988)	(CA 35)/FIS 35	0.169	0.00355	2.100		yes
169	WALLNER	PRL 112, 192501 (2014)	CA 35	99.5	1.99	2.000		yes
170	WALLNER	PRL 112, 192501 (2014)	CA 35	101.5	2.03	2.000		yes
171	ATOYI	Acta Cryst. 20, 587 (1966)	SCR 35	12.3	0.5	4.065		yes
172	WILLIS	PRS A274, 122 (1963)	SCR 35	12.8	1	7.812		yes
174	ARIF	PR A35, 2810 (1987)	SCA 35	14	0.2	1.429		yes
175	CEULEMANS	Divadeenam compilation 1970	SCA 35	14.6	0.5	3.425		yes
176	BAYVAS	Divadeenam compilation 1980	SCA 35	14.3	1.5	10.489		yes
177	FOOTE	Divadeenam compilation 1958	SCR 35	15	1	6.667		yes
178	OLEKSA	Phys Rev 109, 1645 (1958)	SCA 33	13.4	0.5	3.731		yes
179	GREEN	JNM 51, 281 (1974)	SCR 33	12.6	0.7	5.556		yes
180	VERTEBNYI	Divadeenam compilation 1974	SCA 33	13.2	0.6	4.545		yes
181	GREEN	JNM 34, 281 (1970)	SCA 39	7.7	0.6	7.792		yes
182	ATOYI	Acta Cryst. 20, 587 (1966)	SCA 39	7.3	0.7	9.589		yes
183	ROOF	Acta Cryst. 15, 351 (1962)	SCA 39	7.3	0.7	9.589		yes
184	PRONYAEV	Internal analysis 2026-04-17	SCA 41	12	0.8	6.667		yes
185	ANTONOV	Capote 2026-05-04 list	(ABS 35)+SCA 35	695	15	2.158		yes
186	GERASIMOV	Capote 2026-05-04 list	(ABS 35)+SCR 35	692	8	1.156	UPD	yes
187	LEONARD	Capote 2026-05-04 list (1954)	(ABS 35)+SCR 35	701	14	1.997	UPD	yes
188	PSCHENICHNY	Capote 2026-05-04 list (1978)	(ABS 33)+SCR 33	588	9	1.531	UPD	yes
189	HARVEY-1	Capote 2026-05-04 list (1979)	(ABS 33)+SCR 33	588	9	1.531	UPD	yes
192	ANDERSON	Capote 2026-05-04 list (1945)	(ABS 39)+SCR 39	1045	25	2.392		yes
193	PALEVSKY	Capote 2026-05-04 list (1955)	(ABS 39)+SCR 39	1030	10	0.971	UPD	yes
195	TOVESSON	2010 (X4 14271)	(FIS 39)/FIS 35	1.286	0.0257	2.000	NEW	yes
196	GWIN	1984 (X4 12905)	FIS 39	750.2	12.8	1.700	NEW	yes
197	ZHURAVLEV	1976 (X4 40476)	(FIS 39)/FIS 35	1.274	0.0255	2.000	NEW	yes
198	DERUYTTER	1970 (X4 20143)	FIS 39	749.3	3.75	0.500	NEW	yes
199	DERUYTTER	1972 (X4 20132)	FIS 39	748.8	7.49	1.000	NEW	yes
200	STAVISSKII	1965 (X4 41635)	FIS 39	748	7.48	1.000	NEW	yes
201	EGELSTAFF	1955 (X4 23526)	(FIS 41)/FIS 39	1.33	0.0399	3.000	NEW	yes
202	JAFFEY	1955 (X4 12521)	(FIS 41)/FIS 39	1.362	0.0232	1.700	NEW	yes
203	JAMES	1965 (X4 20940)	FIS 41	1014.4	20.3	2.000	NEW	yes
204	WESTON	1978 (X4 10768)	FIS 41	1018.8	27.5	2.700	NEW	yes
205	TOVESSON	2010 (X4 14271)	(FIS 41)/FIS 35	1.73	0.0346	2.000	NEW	yes
206	BERCEANU	1982 (X4 30750)	(FIS 33)/FIS 35	0.892	0.0178	2.000	NEW	yes
207	ZHURAVLEV	1976 (X4 40476)	(FIS 33)/FIS 35	0.908	0.0136	1.500	NEW	yes
208	WENHAI	1975 (X4 32624)	(FIS 33)/FIS 35	0.916	0.00916	1.000	NEW	yes

No.	Author	Reference	Quantity	Value	σ (abs)	σ (%)	Status	In fit
209	DERUYTTER	1974 (X4 20411)	FIS 33	533.2	2.67	0.500	NEW	yes
210	DERUYTTER	1971 (X4 20131)	FIS 35	586.6	4.69	0.800	NEW	yes
211	GWIN	1984 (X4 12905)	FIS 35	586.1	1.76	0.300	NEW	yes
212	BORCEA	1973 (X4 30140)	FIS 35	586.3	7.62	1.300	NEW	yes
213	AMADUCCI	2022 (X4 23453)	FIS 35	586.2	2.93	0.500	NEW	yes
214	DERUYTTER	1970 (X4 20143)	FIS 35	587.6	2.35	0.400	NEW	yes
215	YE	1964 (X4 32827)	FIS 35	582	8.73	1.500	NEW	yes
216	FRAYSSE	1965 (X4 20071)	FIS 35	584.3	8.76	1.500	NEW	yes
217	SAFFORD	1959 (X4 12439)	FIS 35	580	6.96	1.200	NEW	yes
218	MOORE	1960 (X4 12341)	(ABS 33)+SCA 33	587	18	3.066	NEW	yes
219	LYNN	1955 (X4 21027)	(ABS 35)+SCR 35	704	14	1.989	NEW	yes
220	YOUNG	1968 (X4 10406)	(ABS 41)+SCR 41	1390	10	0.719	NEW	yes

Decisions (flags F1–F13)

Each open interpretation point from the 2026-06-23 review, resolved with the most plausible reading and recorded here for confirmation by the working group. Full context in the manifest §F.

F1 — Pattenden 56 Pu-239 uncertainty. The fission table gives "1039(5%)" while the dedicated Pu-239 (n,tot) table gives "1039(25)" (≈2.41%). The later, dedicated (n,tot) revision with the local plot-fit is taken as authoritative → **1039 ± 25 b** (No. 75).

F2 — Leonard 56 Pu-239 (No. 194) — keep or delete. The Pu-239 (n,tot) table shows an adjusted 1042(25), but the fission email and the local analysis both flag it an outlier → **deleted**.

F3 — Lynn 55 U-235 (n,tot) — add or not. Listed as a "NEW set" 704(14) in the consolidated U-235 (n,tot) table, though the local notes advised against adding it (old; large local-vs-integral discrepancy). The consolidated table governs → **added** (No. 219); dissent recorded. Worth confirming.

F4 — Bollinger 52 U-233 (n,tot) — add or not. Present in the local working notes but omitted from the consolidated U-233 (n,tot) list (only Moore carried over as NEW DATA) → **not added**.

F5 — Deruytter 74 FIS(^{233}U) = 533.2 — missing uncertainty. No uncertainty quoted. Assigned **σ = 0.5%** (No. 209), matching the precision of the other Deruytter absolute fission values. **Provisional — confirm the intended value.**

F6 — U-234(n,γ) = 100.9 b — missing uncertainty. Only the central value (B80) was given. The existing relative uncertainty **2.086%** is kept (No. 5); remains a fixed input.

F7 — Watanabe-FIS41 / Deruytter61-FIS35 — update or new. Both match existing entries by author/year/quantity/EXFOR id and are tagged as re-fits → treated as **updates** to Nos. 108 / 98, not new points. The systematic code on No. 98 (B) is retained.

F8 — Berceanu 77 (No. 101) "CONTAINS BORCEA 70". Read as a double-counting warning → **No. 101 kept as-is; Borcea 70 not added separately.**

F9 — Berceanu U-235 FIS35 589.3 "replace by Borcea 73". Interpreted literally → **Berceanu 589.3 dropped; Borcea 73 = 586.3 (1.3%) added** (No. 212).

F10 — Harvey 79 U-233 numbering. The survivor is labelled "data #189" while the quoted old value (592) matches No. 190. Net intent is one Harvey at 588(9) → **No. 189 set to 588(9), No. 190 deleted** (the survivor is overwritten, so the old 585-vs-592 distinction is immaterial).

F11 — Gerasimov 62 U-235 uncertainty. Fission email "692(1.4%)" vs U-235 (n,tot) email "692(8)" (≈1.16%). The dedicated (n,tot) absolute value is used → **692 ± 8 b** (No. 186).

F12 — Shape measurements. Bowman, Melkonian, Schrack, Wagemans 79/88 and Mihailescu are shape (relative-to-thermal) measurements → **not added** as absolute points. (This Melkonian ≠ the U-235 (n,tot) entry No. 63, which is retained.)

F13 — Classification of the new transmission points. "Keep current SCA/SCR" does not cover brand-new entries. Young (No. 220) is explicitly ABS+SCR. **Moore (No. 218) assigned ABS+SCA** (consistent with the local U-233 working assumption) and **Lynn (No. 219) assigned ABS+SCR** (era convention) — both provisional. Note: this only affects the favoured Step-5-style fit; under a Step-6-style "all ABS+SCR → ABS+SCA" conversion the choice is moot.

Files Produced (Round 3)

- `axton1986/axton1986_data_capote_round3.json` — round-3 database (205 measurements), regenerated by `evaluation/build_round3_json.py`
- `evaluation/build_round3_json.py` — auditable generator (applies the manifest to the Step-5 base)
- `evaluation/benchmark_capote_round3.py` — round-3 fit benchmark
- `evaluation/build_round3_report.py` — this report generator
- `evaluation/round3_summary.json` — machine-readable round-3 results (+ Step-5 reference)
- `references/emails/tnc_update_roberto_20260623/` — paraphrased discussions + datapoint-level manifest with decision register